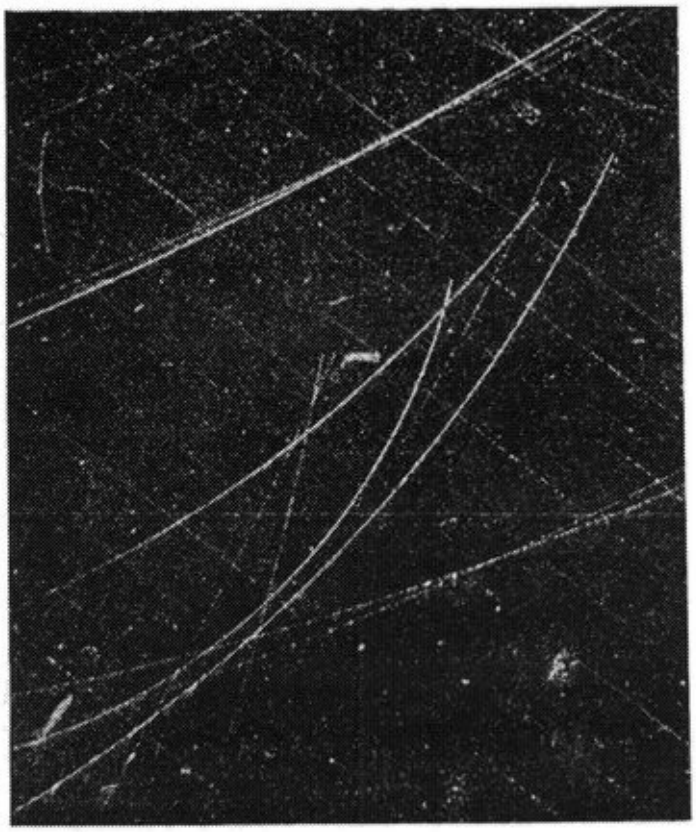


Figure 17A.—Scattering of neutrons in hydrogen.

The situation is, however, not quite so simple as we have made it appear. If the analogy of nuclear and electric forces were as we have been supposing, we would be in a position to determine the frequency of the occurrence of beta decay in a manner similar to that employed to determine the frequency of the occurrence of the emission of a photon in the extranuclear structure of the atom. When an extranuclear atomic structure is in an excited state, there exists in it at any given moment a certain probability of the emission of a light ray. By ‘certain probability’ we mean the following: In the wave aspect, the continuous movement of electrons causes a wave radiation to issue forth. In the particle aspect, there exists at any given moment a certain probability of the emission of a photon. These two views of the situation are linked to each other by the fact that the probability of radiation is given by the intensity of the emitted wave. The stronger is the wave, the greater is the probability of radiation, and the shorter-lived is the excited state. The duration of the excited state depends, therefore, on the amplitude of the vibration of the electrons.

The lifetime of a beta-unstable nucleus thus depends on the intensity of the wave radiation issuing forth from it. But if we carry out this computation on the ground of the considerations outlined above, we arrive at lifetimes much shorter than those actually observed. There still exists a discrepancy at this point, and this realization led the Japanese scientist Yukawa to a somewhat